

FineBadminton: A Multi-Level Dataset for Fine-Grained Badminton Video Understanding

Xusheng He
xsxshxs22@gmail.com
Harbin Institute of Technology,
Shenzhen

Wei Liu
liuwei030224@gmail.com
Harbin Institute of Technology,
Shenzhen

Shanshan Ma*
mass1@cesi.cn
China Electronics Standardization
Institute

Qian Liu
leonaliuqian@gmail.com
Shandong University

Chenghao Ma
mach@cesi.cn
China Electronics Standardization
Institute

Jianlong Wu*
wujianlong@hit.edu.cn
Harbin Institute of Technology,
Shenzhen

Abstract

Fine-grained analysis of complex and high-speed sports like badminton presents a significant challenge for Multimodal Large Language Models (MLLMs), despite their notable advancements in general video understanding. This difficulty arises primarily from the scarcity of datasets with sufficiently rich and domain-specific annotations. To bridge this gap, we introduce FineBadminton, a novel and large-scale dataset featuring a unique multi-level semantic annotation hierarchy (Foundational Actions, Tactical Semantics, and Decision Evaluation) for comprehensive badminton understanding. The construction of FineBadminton is powered by an innovative annotation pipeline that synergistically combines MLLM-generated proposals with human refinement. We also present FBBench, a challenging benchmark derived from FineBadminton, to rigorously evaluate MLLMs on nuanced spatio-temporal reasoning and tactical comprehension. Together, FineBadminton and FBBench provide a crucial ecosystem to catalyze research in fine-grained video understanding and advance the development of MLLMs in sports intelligence. Furthermore, we propose an optimized baseline approach incorporating Hit-Centric Keyframe Selection to focus on pivotal moments and Coordinate-Guided Condensation to distill salient visual information. The results on FBBench reveal that while current MLLMs still face significant challenges in deep sports video analysis, our proposed strategies nonetheless achieve substantial performance gains. The project homepage is available at <https://finebadminton.github.io/FineBadminton/>.

CCS Concepts

• **Computing methodologies** → **Computer vision**; • **Information systems** → **Multimedia streaming**.

*Corresponding authors.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
MM '25, Dublin, Ireland.

© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-2035-2/2025/10
<https://doi.org/10.1145/3746027.3758218>

Keywords

Sports Understanding, Multi-modal LLMs, Badminton Video Dataset

ACM Reference Format:

Xusheng He, Wei Liu, Shanshan Ma, Qian Liu, Chenghao Ma, and Jianlong Wu. 2025. FineBadminton: A Multi-Level Dataset for Fine-Grained Badminton Video Understanding. In *Proceedings of the 33rd ACM International Conference on Multimedia (MM '25)*, October 27–31, 2025, Dublin, Ireland. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3746027.3758218>

1 Introduction

Multimodal Large Language Models (MLLMs) have demonstrated remarkable progress in general video understanding tasks [7, 14, 19, 24, 40]. However, their capabilities are significantly challenged in scenarios demanding fine-grained analysis [4, 11, 21, 22, 32], where subtle spatio-temporal details, complex event interactions, and domain-specific knowledge are paramount. This limitation is particularly pronounced in intricate sports like badminton, which is characterized by high-speed rallies, a diverse set of nuanced strokes executed with split-second timing, and rapid tactical decision-making [3, 12, 25]. These complexities push the boundaries of current video analysis far beyond coarse action recognition, requiring deeper and hierarchical gameplay comprehension [28, 29, 39].

A critical bottleneck hindering advancement in this area is the scarcity of richly annotated datasets specifically designed for fine-grained badminton analysis. While general video datasets and some sports-specific collections exist, they often lack the necessary granularity or domain relevance. Although existing badminton resources [20, 37, 38] are valuable for foundational tasks like stroke classification or shuttlecock tracking, they typically focus on more basic aspects and do not capture the deeper tactical intentions, precise execution nuances, or evaluative dimensions essential for comprehensive understanding. For instance, they may identify a "net shot" but fail to distinguish a strategically deceptive cross-court spinning net shot from a standard one, or to assess its impact on the rally. Manually annotating the dense, complex, and multi-layered events inherent in badminton at the necessary scale and detail is prohibitively expensive and prone to inconsistencies.

To bridge this critical gap, we introduce **FineBadminton**, a novel and large-scale dataset carefully designed for fine-grained badminton video understanding. FineBadminton features a unique multi-level annotation hierarchy providing unprecedented depth: (1) **Foundational Actions**, offering detailed classifications of stroke



Figure 1: An annotation example from FineBadminton illustrating its multi-level hierarchy. The figure showcases annotations for distinct shot moments within a single rally, detailing (1) Foundational Actions, (2) Tactical Semantics, and (3) Decision Evaluation.

types and their execution subtleties; (2) **Tactical Semantics**, which describe ball trajectory dynamics and infer player strategic intentions, like deceptive plays or attacking formations; and (3) **Decision Evaluation**, providing expert-informed assessments of individual shot quality and entire rally narratives, including point-winning and error-inducing sequences. This hierarchical structure, exemplified in Figure 1, facilitates a nuanced analysis by systematically detailing foundational actions, interpreting their tactical significance, and evaluating their contribution to the rally’s outcome.

Recognizing the impracticality of purely manual annotation for such intricate detail, FineBadminton is developed using an innovative MLLM-driven automated annotation pipeline. This pipeline synergistically integrates foundational video models for structural parsing (e.g., player and ball tracking, hit event detection) with the advanced reasoning capabilities of MLLMs and LLMs to propose detailed and multi-level annotations. These proposals are subsequently refined by human experts, ensuring high-quality annotations with professional badminton insight. This human-in-the-loop approach significantly accelerates the creation of complex and hierarchical annotations while maintaining domain accuracy.

To validate FineBadminton’s utility and promote further research, we establish **FBench**, a comprehensive benchmark designed to evaluate MLLM capabilities in nuanced and tactical sports video understanding. FBench features a hierarchical task structure including Count, Action, Position and Cognition categories, assessing a model’s capabilities in complex temporal reasoning, fine-grained action understanding, spatial awareness, and strategic inference within dynamic rally contexts. Furthermore, we propose an optimized baseline approach, underpinned by two novel badminton-specific strategies: Hit-Centric Keyframe Selection and Coordinate-Guided Information Condensation. This approach is designed to effectively adapt MLLMs to the demanding fine-grained tasks benchmarked by FBench. We validate the effectiveness of these strategies through extensive experiments on FBench, showing performance improvements across leading open-source and proprietary commercial MLLMs [1, 27, 33, 34, 47].

Our contributions can be summarized as follows:

- (1) We introduce FineBadminton, a large-scale dataset of 3,215 rally clips (33,325 strokes), featuring an unparalleled multi-level semantic annotation hierarchy for in-depth badminton analysis beyond simple stroke recognition.
- (2) We develop an innovative MLLM-driven Annotation Pipeline enabling scalable and consistent creation of fine-grained, hierarchical annotations with expert refinement.
- (3) Building upon FineBadminton, we present FBench benchmark, designed with structured tasks for rigorous evaluation of nuanced sports video understanding.
- (4) We propose an optimized approach with novel strategies to effectively adapt MLLMs for fine-grained analysis on FBench.

2 Related Work

2.1 Video Understanding Datasets

Video understanding datasets have rapidly evolved from early benchmarks focused on video classification [15, 31] and action recognition [5, 10], to encompassing more complex tasks such as video question answering [17, 44]. Recent advancements have further driven the field towards finer-grained understanding, with datasets targeting sophisticated temporal reasoning [16, 19, 26, 41] and the analysis of long-form videos [2, 6, 46]. However, while these general-purpose datasets significantly advance video intelligence, their application to specialized domains like sports is often constrained by limited domain-specific content and coarse-grained annotations, which lack the nuanced details crucial for in-depth analysis. These limitations highlight the need for fine-grained datasets tailored to complex and dynamic domains like sports.

2.2 Sports Video Understanding Datasets

Sports video understanding has emerged as a rapidly advancing research area, driven by its significant practical applications. Existing datasets in this domain can be broadly categorized into multi-sport collections [18, 42, 43] and sport-specific resources [12, 13, 25, 28, 29, 45, 48]. Representative multi-sport datasets include Sports-QA [18] which encompasses eight sports using videos from action recognition datasets MultiSports [23] and FineGym [30]. In contrast, sport-specific datasets offer more specific and fine-grained annotations, such as SoccerReplay-1988 [28] and SoccerNet-XFoul [13] for soccer analysis, and FineSports [45] targeting basketball activities. In this work, we focus on badminton. Existing badminton datasets [20, 37, 38] provide foundational stroke-level annotations, yet they lack the granularity, tactical semantics, and evaluative depth necessary for comprehensive understanding.

To address these limitations, we introduce **FineBadminton**, featuring a multi-level annotation framework for badminton analysis, alongside **FBench**, a comprehensive benchmark.

3 FineBadminton: A Fine-Grained Badminton Video Dataset with Hierarchical Semantics

To facilitate in-depth, fine-grained badminton gameplay analysis, we introduce **FineBadminton**, a novel large-scale dataset with comprehensive multi-level semantic annotations. This section outlines its construction methodology (Section 3.1), MLLM-driven annotation pipeline (Section 3.2), and statistical analysis (Section 3.3).

3.1 Dataset Construction

FineBadminton is meticulously constructed through several key stages, beginning with data sourcing and culminating in comprehensive quality control, to ensure a reliable and high-quality dataset.

Data Preparation. FineBadminton is curated from publicly accessible professional badminton match videos, officially released by the Badminton World Federation (BWF) on YouTube, covering events like the Olympics, World Championships, and various Open series events. The predominant camera perspective is an overhead view from behind the court, supplemented by a smaller subset of side-angle clips. All video content features a resolution of 720p or higher, with a consistent frame rate of 25 FPS.

Annotation Scheme. To capture badminton’s intricate semantic complexity, we design a multi-level hierarchical annotation scheme. This structure facilitates a progressive and comprehensive understanding of sports semantics across three abstraction levels: Foundational Actions, Tactical Semantics, and Decision Evaluation. All terminology and classifications within these levels are developed in collaboration with professional badminton players to ensure domain accuracy and relevance.

The **Foundational Actions** level focuses on the granular classification of individual badminton strokes, identifying the specific type of shot. Building upon the ShuttleSet22 framework and expert input, we deconstruct single strokes into the following 11 primary categories. These primary categories are further refined into 20 sub-categories based on the nuances of hand and wrist movements during execution. For instance, a ‘smash’ can be sub-classified as a ‘jump smash’ or a ‘slice smash’. Notably, a single stroke can belong to multiple sub-categories, enhancing descriptive granularity.

The **Tactical Semantics** level incorporates ball movement dynamics and inferred player intent, connecting physical actions to tactical awareness. The lexicon for this level, derived from common parlance in expert commentary and validated by professional players, includes distinctions based on trajectory (e.g., ‘straight line,’ ‘cross-court’), position relative to the opponent (e.g., ‘passing shot,’ ‘body shot,’ ‘overhead shot’), and outcome relative to the court (e.g., ‘out of bounds,’ ‘hitting into the net’). Player intent reflects subjective elements such as ‘deception,’ or situational assessments like ‘defensive play’ and ‘passive/transitional shot.’

The **Decision Evaluation** level provides a higher-level and temporally extended assessment, evaluating individual shot quality and the overall rally’s narrative. This level incorporates quality scores and textual descriptions for each shot. It broadens the perspective to an entire rally, by analyzing its characteristics, identifying key shots, and detailing reasons for points won or lost. It also captures difficult-to-categorize in-game events (e.g., distinctive postures, critical movements) to ensure a holistic semantic representation.

Annotation Process. Given the complexity and hierarchical nature of our requirements, we develop custom GUI-based annotation software using Python. The annotation proceeds in two principal phases: (1) An initial phase relying entirely on manual annotation by trained human annotators. (2) A second phase where human annotators review and refine outputs generated by our automated annotation pipeline (detailed in Section 3.2). This approach allows us to scale the annotation effort while maintaining high quality.

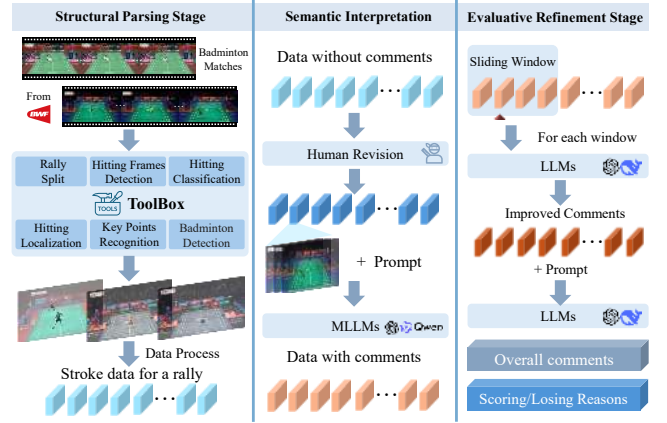


Figure 2: Architecture of our automated annotation pipeline. The three-stage process first uses Structural Parsing to extract structured data from video, then Semantic Interpretation to generate descriptions for each stroke, and finally Evaluative Refinement to produce a cohesive rally evaluation.

Quality Control. A stringent quality control protocol ensures the accuracy, consistency, and reliability of FineBadminton. The annotation team, comprising four experienced badminton enthusiasts, undergoes comprehensive training and completes pilot exercises before starting the main task. In the initial manual phase, each data sample is independently annotated by two individuals; a senior annotator then reviews these annotations, resolves discrepancies, and consolidates them into a unified and high-quality label. For the subsequent pipeline-assisted phase, outputs from the automated system are thoroughly reviewed and refined by two human annotators to ensure compliance with our established quality standards.

3.2 Automated Annotation Pipeline

To efficiently construct the FineBadminton dataset with rich and hierarchical annotations, we develop a multi-stage automated annotation pipeline, illustrated in Figure 2. The pipeline processes raw badminton match videos to extract structured information, interpret it semantically, and refine these interpretations into nuanced evaluations. The pipeline comprises three core stages: Structural Parsing, Semantic Interpretation, and Evaluative Refinement.

The initial **Structural Parsing Stage** deconstructs raw match footage into fundamental spatio-temporal events and attributes. This stage employs a “toolbox” of specialized models to extract key information crucial for badminton analysis. For ball trajectory detection, we employ TrackNetV3[3] due to its robustness against occlusion. Court boundary and player keypoint localization are achieved using FastRCNN[8]. For identifying hit events (frames where a stroke occurs) and enabling deeper action understanding, we utilize VideoMAE[35, 36], fine-tuned on badminton videos, as the visual feature extractor. These features, extracted from frames around the detected hits, are then combined with corresponding ball and player coordinate data and fed into a spatio-temporal feature learning and fusion module. This module enables task-specific prediction heads to classify stroke types and localize shuttlecock landing spots. The resulting structured data $\mathcal{D}$, containing elements like ball trajectories, player/court localizations, hit instances, stroke

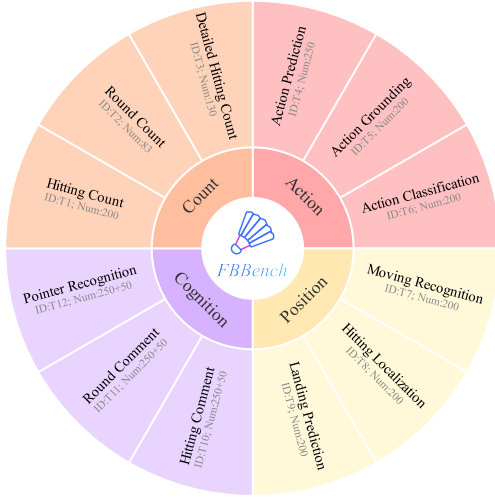


Figure 3: Hierarchical task structure of FBBench, showcasing the four primary categories (Count, Action, Position, Cognition) and their respective three sub-categories designed to probe diverse video understanding capabilities.

types, and landing spots, then proceeds to manual verification, as detailed in Section 3.1.

The second stage, **Semantic Interpretation**, translates the extracted structured data into rich textual narratives. This phase harnesses the advanced cross-modal reasoning capabilities of MLLMs. For each j -th identified stroke in a rally, its visual context I_j (three frames sampled around the hit instance) and associated structured data D_j are fed into the MLLM. The model then generates a fine-grained textual description T_j for that stroke, characterizing and evaluating its specific qualities.

Finally, the **Evaluative Refinement Stage** produces contextually aware assessments of rally dynamics. While the previous stage provides detailed descriptions T_j for each stroke, these outputs may lack rally context. To address this, a refining LLM processes each T_j . This refinement considers a contextual window composed of a total of w stroke descriptions, typically centered around the current stroke T_j , producing an improved T'_j . Subsequently, for a rally R comprising a sequence of N refined stroke descriptions ($T'_1, T'_2, \dots, T'_N$), a comprehensive overall rally evaluation and reasons for scoring or losing is generated by a summarizing LLM. This final step ensures that the interplay between shots and the evolving tactical situation is more accurately captured in the narrative.

3.3 Dataset Analysis

The FineBadminton dataset is constructed from 120 singles matches over the past five years. These matches are segmented into 3,215 rally clips, totaling 33,325 strokes. The average duration of each rally is 12.4 seconds. Notably, FineBadminton is, to our knowledge, the first public badminton dataset that features such a detailed multi-level hierarchical annotation scheme combined with rich, free-form textual descriptions for strokes and rallies.

The dataset's hierarchical annotation provides a granular breakdown of badminton gameplay. At the Foundational Actions level, there are 11 primary hit types and 20 subtypes. The shuttlecock's position on the court is classified into 9 distinct regions.

The Tactical Semantics level further enriches the dataset with 3 categories of player actions, 9 strategic classifications, and 6 types of shot characteristics. The outcome of last hit has 3 classes.

In the Decision Evaluation level, individual shots are assigned a quality score from 1 to 7, with hitting comments averaging 37.4 words. The round-level evaluation includes an "Overall Evaluation" averaging 63 words, "Scoring Reason" averaging 41 words, and "Losing Reason" averaging 44 words, providing a comprehensive narrative of the rally's progression and outcome.

4 FBBench: A Multi-Faceted Benchmark

To facilitate a more comprehensive evaluation of badminton video understanding models, we propose **FBBench**, a novel and challenging benchmark. Built upon the meticulously annotated FineBadminton dataset, FBBench is designed to rigorously assess a model's capabilities in deciphering the intricate dynamics of badminton play, pushing the boundaries beyond simple action recognition.

Specifically, inspired by MVBench [19], FBBench emphasizes temporal understanding in fine-grained video analysis, with tasks deliberately crafted to require multi-frame reasoning. FBBench comprises four core domains: **Count**, **Action**, **Position**, and **Cognition**. Each is further divided into three specialized subcategories, culminating in the 12 distinct task types illustrated in Figure 3. This structure ensures a broad and deep evaluation of model capabilities.

Count tasks assess a model's ability to quantify discrete events, such as strokes per rally or rallies per game, sometimes requiring badminton-specific knowledge like distinguishing shot types.

Action tasks focus on fine-grained motor and tactical recognition, involving classifying strokes, predicting plausible returns, or temporally localizing key actions within a rally.

Position tasks target spatio-temporal reasoning, including player localization, movement tracking, and shuttlecock landing prediction based on trajectory and context.

Cognition tasks assess higher-level reasoning, challenging models with tasks such as shot quality evaluation, strategic performance inference, and rally outcome determination, all of which require causal inference from extended evidence.

FBBench has 2,563 question-answer pairs. The majority (2,413) are multiple-choice, designed for objective, scalable evaluation, featuring rule-based or LLM-generated plausible distractors and answer length supervision to enhance robustness. Complementing these, 150 open-ended QA tasks assess nuanced descriptive and analytical capabilities. These are evenly distributed across three subcategories of the Cognition task, requiring free-form textual responses for detailed evaluations, scoring or losing reason analysis.

This dual-format strategy, combined with diverse task categories spanning a difficulty gradient—from fundamental perception (e.g., counting, localization) through game mechanics understanding to advanced semantic parsing and strategic reasoning—establishes FBBench as a comprehensive spectrum of challenges. Consequently, FBBench provides a robust platform for evaluating and advancing video understanding models in sports analysis.

5 Optimized Baseline with Novel Strategies

Given the substantial inter-frame redundancy in badminton videos, which can obscure core actions and reduce MLLM efficiency, a

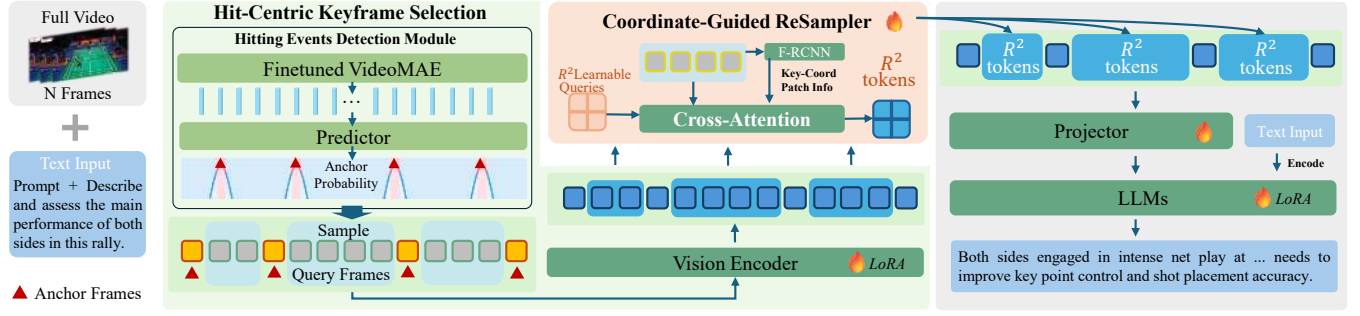


Figure 4: Architecture of the optimized baseline approach. It employs a two-stage strategy: Hit-Centric Keyframe Selection and Coordinate-Guided Visual Information Condensation, extracting salient information from badminton videos.

transformation that focuses on salient action segments to create a compact representation is essential. Our optimized baseline approach centers on such an efficient transformation, denoted as $\mathcal{R}$, which converts a raw video V into a compact token sequence T . This representation T is then fed to an LLM along with a textual query q to generate a response r : $r = \text{LLM}(T, q)$. The transformation $\mathcal{R}$ is specifically designed to address the high temporal density of actions and visual redundancy inherent in badminton videos.

As illustrated in Figure 4, $\mathcal{R}$ comprises a carefully designed two-stage process: $\mathcal{R}(V) = C(S(V))$. Keyframe Selection (S) firstly pinpoints action-rich pivotal moments to mitigate temporal redundancy. Visual Information Condensation (C) then extracts salient cues from these selected frames, yielding a compact input.

5.1 Hit-Centric Keyframe Selection

The Keyframe Selection stage (S) identifies salient temporal segments. Its goal is to extract a sparse yet representative set of frames capturing tactically significant moments, critical for comprehensive game understanding.

Our hit-centric strategy S leverages a hit-event detection module, adapted from the annotation pipeline, utilizing the same VideoMAE model to extract per-frame visual features. These features are subsequently processed through a projection module, which comprises an attention pooling layer and a transformer-based temporal encoder—to output a hit probability $P(\text{hit}|f_i, V)$ for each frame f_i in video segment V . Hitting frames are designated as **anchor frames** ($\mathcal{A}$), representing significant action points.

These M anchor frames, $\mathcal{A} = \{A_1, \dots, A_M\}$, partition the video into $M - 1$ inter-hit segments. To capture crucial pre- and post-hit dynamics, we sample a variable number of **query frames** from each segment. The total number of selected frames is $M + \sum_{i=1}^{M-1} N_i$, where N_i is the number of query frames in the i -th segment.

All selected frames are processed by the MLLM’s visual encoder. Each frame is transformed into K_{enc} visual tokens. The complete set of these initial frame token embeddings is denoted as K_{emb} .

5.2 Coordinate-Guided Condensation

The Visual Information Condensation stage (C) distills K_{emb} into the compact token sequence $T = C(K_{emb})$, to minimize static background redundancy while preserving crucial semantic cues.

Anchor frame tokens E_{A_j} are preserved. For query frame tokens $X_{i:i+1}$, we employ a ReSampler module, inspired by the Perceiver architecture. This module uses R learnable queries $Q_{learn} \in \mathbb{R}^{R \times D}$ to distill information via cross-attention.

This resampling is **coordinate-guided**. We detect salient game elements (ball, players, court keypoints) in query frames using a FastRCNN-based object detector. These spatial detections guide the ReSampler. Let q' be processed learnable queries, and $k^{(i)}, v^{(i)}$ be key and value sequences for the i -th segment, derived from Q_{learn} and $X_{i:i+1}$ respectively. The attention scores for segment i are:

$$\text{Score}^{(i)} = \frac{q'(k^{(i)})^T}{\sqrt{D}} + B_{\text{coord}}^{(i)}$$

where D is the dimension of key vectors. The coordinate-bias tensor $B_{\text{coord}}^{(i)} \in \mathbb{R}^{B \times R \times (N_i \cdot K_{enc})}$ has element (b, p, q) set to a scalar hyperparameter $\alpha > 0$ if the q -th input token in $X_{i:i+1}$ spatially aligns with a detected key game element, and 0 otherwise. This additive bias effectively steers the ReSampler’s attention towards these critical visual regions. The condensed representation for the query segment is $R_{i:i+1} = \text{Softmax}(\text{Score}^{(i)})v^{(i)} \in \mathbb{R}^{B \times R \times D}$.

The final token sequence T concatenates anchor frame tokens and condensed query segment representations temporally:

$$T = \text{Concat}(E_{A_1}, R_{1:2}, E_{A_2}, \dots, R_{M-1:M}, E_{A_M})$$

This results in $T \in \mathbb{R}^{B \times (M \cdot K_{enc} + (M-1) \cdot R) \times D}$, emphasizing pivotal hits and salient inter-hit cues for LLM processing.

Fine-Tuning. The training set consists of approximately 60,000 QA pairs. We construct these pairs from the FineBadminton, designing question types based on the categories outlined in the FBBench. This results in 10% multiple-choice questions and the remainder open-ended questions, all of which are formulated and trained as standard sequence-to-sequence generation tasks for the LLM. Within the MLLM, the visual encoder and the LLM were fine-tuned using LoRA, whereas the ReSampler and the hit-centric projection module underwent full-parameter training.

6 Experiments

6.1 Experimental Settings

Models and Methodologies. We evaluated several state-of-the-art models on our FBBench. These include prominent commercial models such as Gemini 2.5 Pro[34], Doubao 1.5 Pro[33], and GPT-4.1[27], as well as leading open-source models such as Qwen2.5VL-7B[1], and VideoLLaMA3-7B[47]. All models were benchmarked using two distinct video frame processing approaches: (1) a baseline uniform sampling at 2 FPS, and (2) our proposed S strategy.

Experimental Details. For the commercial models, accessed via APIs, we evaluated the impact of our S strategy. For the open-source

Table 1: Quantitative Comparisons on FBBench. Numbers indicate correctly answered questions. For open-ended questions (marked with *), the number represents total score (0–10 per response). *S* and *C* denote Hit-Centric Keyframe Selection and Coordinate-Guided Condensation in Section 5. *T_i* denotes the *i*-th type of question, refer to Figure 3 for the detailed mapping.

Model	Count			Action			Position			Cognition						Overall	
	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T10*	T11	T11*	T12	T12*	Choice	Open-ended
Full Score	200	83	130	250	200	200	200	200	200	250	500	250	500	250	500	2413	1500
Commerical Models																	
Gemini 2.5 Pro	69	33	34	67	70	44	54	63	45	124	192	114	214	155	325	872 (36.14%)	731 (48.73%)
Gemini 2.5 Pro + <i>S</i>	79	28	35	68	71	54	58	70	53	131	202	117	222	168	335	932 (38.62%)	759 (50.60%)
Doubao 1.5 pro	71	23	45	55	50	36	40	61	52	133	165	112	213	133	286	811 (33.61%)	664 (44.27%)
Doubao 1.5 pro + <i>S</i>	71	30	48	56	57	39	48	68	45	130	188	121	227	143	294	856 (35.47%)	709 (47.27%)
GPT-4.1	47	12	40	76	55	60	40	30	48	110	183	98	217	128	310	743 (30.79%)	710 (47.33%)
GPT-4.1 + <i>S</i>	46	17	41	70	61	64	44	39	49	112	186	100	213	138	316	782 (32.41%)	715 (47.67%)
Open-Source Models																	
Qwen2.5VL-7B	48	19	30	50	39	33	52	41	41	60	110	62	79	82	46	557 (23.08%)	235 (15.67%)
Qwen2.5VL-7B + <i>S</i> + <i>C</i>	70	24	36	92	44	87	56	55	98	170	172	142	205	141	278	1015 (42.06%)	655 (43.67%)
VideoLLaMA3-7B	42	10	20	17	44	24	30	20	12	45	95	55	66	40	47	359 (14.88%)	208 (13.87%)
VideoLLaMA3-7B + <i>S</i> + <i>C</i>	52	24	25	85	54	80	40	68	96	165	167	130	198	135	264	954 (39.54%)	629 (41.93%)

models, we applied our optimized baseline incorporating both the *S* and *C* strategies, and trained these models on 4 A100-40G GPUs. To match the frame count of the uniform baseline, uniformly sampled inter-keyframe frames were added when *S* yielded fewer frames.

Metrics. Performance on multiple-choice questions was measured by accuracy, calculated based on the total number of correct answers. To mitigate potential biases from option ordering, we implemented a re-evaluation loop where options were shuffled for each assessment. For open-ended questions, we utilized GPT-4.1 as an evaluator. Each response was scored by GPT-4.1 on a scale of 0 to 10 points, based on relevance, correctness, and completeness, against a reference answer. The overall performance for open-ended tasks was reported as the cumulative score across all such questions.

6.2 Quantitative Evaluation

Quantitative results on FBBench are presented in Table 1. Among commercial models, performance rankings aligned with general expectations: Gemini 2.5 Pro led, followed by Doubao 1.5 Pro, and then GPT-4.1. The benchmark’s difficulty is evident, as even the top-performing Gemini 2.5 Pro (with *S*) achieved an overall multiple-choice score of 932 (representing 38.6% accuracy), indicating significant headroom for improvement. Across all models, tasks requiring deep domain-specific knowledge (e.g., T3, T6), subtle action discrimination (e.g., T1, T7), or complex reasoning (e.g., T10-T12) proved particularly challenging. Notably, our strategy *S* consistently enhanced the performance of these commercial models over uniform sampling with an equivalent frame count.

Remarkably, the smaller 7B open-source models, which initially performed below 25% overall accuracy, demonstrated substantial improvements after fine-tuning with *S* and *C* strategies. The optimized Qwen2.5VL-7B achieved an overall 42.06% multiple-choice accuracy, surpassing all evaluated commercial models including the leading Gemini 2.5 Pro (with *S*). Furthermore, their performance on open-ended QA tasks (e.g., Qwen2.5VL-7B: 655) also became highly competitive, approaching the levels of top commercial models.

Table 2: Ablations on FBBench Multi-Choice QA. *FT* denotes finetuning. The **blue background indicates our approach.**

<i>S</i>	<i>C</i>	<i>FT</i>	Count	Action	Localization	Cognition	Overall
✗	✗	✗	72	85	62	140	359
✗	✗	✓	70	132	124	301	627
✓	✗	✓	75	153	147	362	737
✗	✓	✓	81	195	186	394	856
✓	✓	✓	101	219	204	430	954

6.3 Ablation Studies

To isolate and validate the effectiveness of our proposed *S* and *C* strategies, we conducted ablation studies on VideoLLaMA3. The results are presented in Table 2.

Fine-tuning VideoLLaMA3 on our FineBadminton subset substantially boosted its overall score from 359 to 627, highlighting the critical role of domain-specific fine-tuning. Further enhancements were achieved by integrating our proposed strategies: the *S* strategy (with *FT*) increased the score to 737, and the *C* strategy (with *FT*) reached 856. Notably, our full model, combining all components, achieved the highest score of 954, demonstrating the synergistic benefits of domain-specific fine-tuning with our tailored video processing for effective fine-grained analysis.

7 Conclusion

In this paper, we introduced **FineBadminton**, a novel, large-scale dataset for fine-grained badminton video understanding, distinguished by its comprehensive multi-level semantic annotation hierarchy. The dataset’s construction is supported by an MLLM-driven annotation pipeline with expert refinement. We also presented **FBBench**, a challenging benchmark for rigorous assessment of nuanced video understanding. While our video processing strategies enhance MLLM performance, evaluations on FBBench demonstrate current models struggle with the complex dynamics and tactics of high-speed sports videos. We believe this work provides a crucial foundation to spur advancements in fine-grained video analysis.

Acknowledgments

This work was supported in part by the National Natural Science Foundation of China under Grant 62376069, in part by the Young Elite Scientists Sponsorship Program by CAST under Grant 2023QNRC001, in part by Guangdong Basic and Applied Basic Research Foundation under Grant 2024A1515012027, in part by the Shenzhen Science and Technology Program under Grant KQTD2024 0729102207002 and Grant ZDSYS20230626091203008, and in part by Jiangsu Science and Technology Major Program under Grant BG2024041.

References

- [1] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibao Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, et al. 2025. Qwen2. 5-vl technical report. *arXiv preprint arXiv:2502.13923* (2025).
- [2] Lin Chen, Xilin Wei, Jinsong Li, Xiaoyi Dong, Pan Zhang, Yuhang Zang, Zehui Chen, Haodong Duan, Zhenyu Tang, Li Yuan, et al. 2024. Sharegpt4video: Improving video understanding and generation with better captions. In *Proceedings of the Advances in Neural Information Processing Systems*, Vol. 37. 19472–19495.
- [3] Yu-Jou Chen and Yu-Shuen Wang. 2023. Tracknetv3: Enhancing shuttlecock tracking with augmentations and trajectory rectification. In *Proceedings of ACM International Conference on Multimedia in Asia*. 1–7.
- [4] Muhe Ding, Yang Ma, Pengda Qin, Jianlong Wu, Yuhong Li, and Liqiang Nie. 2025. RA-BLIP: Multimodal Adaptive Retrieval-Augmented Bootstrapping Language-Image Pre-training. *IEEE Transactions on Multimedia* (2025).
- [5] Bernard Ghanem Fabian Caba Heilbron, Victor Escorcia and Juan Carlos Nibbles. 2015. ActivityNet: A Large-Scale Video Benchmark for Human Activity Understanding. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. IEEE, 961–970.
- [6] Miquel Farré, Andi Marafioti, Lewis Tunstall, Leandro Von Werra, and Thomas Wolf. 2024. FineVideo. <https://huggingface.co/datasets/HuggingFaceFV/finevideo>.
- [7] Chaoyou Fu, Yuhang Dai, Yongdong Luo, Lei Li, Shuhuai Ren, Renrui Zhang, Zihan Wang, Chenyu Zhou, Yunhang Shen, Mengdan Zhang, et al. 2024. Video-mme: The first-ever comprehensive evaluation benchmark of multi-modal llms in video analysis. *arXiv preprint arXiv:2405.21075* (2024).
- [8] Ross Girshick. 2015. Fast r-cnn. In *Proceedings of the IEEE International Conference on Computer Vision*. IEEE, 1440–1448.
- [9] Team GLM, Aohan Zeng, Bin Xu, Bowen Wang, Chenhui Zhang, Da Yin, Dan Zhang, Diego Rojas, Guanyu Feng, Hanlin Zhao, et al. 2024. ChatGLM: A Family of Large Language Models from GLM-130B to GLM-4 All Tools. *arXiv preprint arXiv:2406.12793* (2024).
- [10] Raghav Goyal, Samira Ebrahimi Kahou, Vincent Michalski, Joanna Materzynska, Susanne Westphal, Heuna Kim, Valentin Haenel, Ingo Fruend, Peter Yianilos, Moritz Mueller-Freitag, et al. 2017. The "something something" video database for learning and evaluating visual common sense. In *Proceedings of the IEEE International Conference on Computer Vision*. IEEE, 5842–5850.
- [11] Hulingxiao He, Geng Li, Zijun Geng, Jinglin Xu, and Yuxin Peng. 2025. Analyzing and Boosting the Power of Fine-Grained Visual Recognition for Multi-modal Large Language Models. *arXiv preprint arXiv:2501.15140* (2025).
- [12] Yuchen He, Zeqing Yuan, Yihong Wu, Liqi Cheng, Dazhen Deng, and Yingcai Wu. 2024. ViSTec: Video Modeling for Sports Technique Recognition and Tactical Analysis. In *Proceedings of the AAAI Conference on Artificial Intelligence*. 8490–8498.
- [13] Jan Held, Hani Itani, Anthony Cioppa, Silvio Giancola, Bernard Ghanem, and Marc Van Droogenbroeck. 2024. X-vars: Introducing explainability in football refereeing with multi-modal large language models. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. IEEE, 3267–3279.
- [14] Kairui Hu, Penghao Wu, Fanyi Pu, Wang Xiao, Yuanhan Zhang, Xiang Yue, Bo Li, and Ziwei Liu. 2025. Video-MMMU: Evaluating Knowledge Acquisition from Multi-Discipline Professional Videos. *arXiv preprint arXiv:2501.13826* (2025).
- [15] Will Kay, Joao Carreira, Karen Simonyan, Brian Zhang, Chloe Hillier, Sudheendra Vijayanarasimhan, Fabio Viola, Tim Green, Trevor Back, Paul Natsev, et al. 2017. The kinetics human action video dataset. *arXiv preprint arXiv:1705.06950* (2017).
- [16] Jie Lei, Tamara L Berg, and Mohit Bansal. 2021. Detecting moments and highlights in videos via natural language queries. In *Proceedings of the Advances in Neural Information Processing Systems*, Vol. 34. NeurIPS Foundation, 11846–11858.
- [17] Jie Lei, Licheng Yu, Mohit Bansal, and Tamara L Berg. 2018. Tvqa: Localized, compositional video question answering. *arXiv preprint arXiv:1809.01696* (2018).
- [18] Haopeng Li, Andong Deng, Qihong Ke, Jun Liu, Hossein Rahmani, Yulan Guo, Bernt Schiele, and Chen Chen. 2024. Sports-qa: A large-scale video question answering benchmark for complex and professional sports. *arXiv preprint arXiv:2401.01505* (2024).
- [19] Kunchang Li, Yali Wang, Yanan He, Yizhuo Li, Yi Wang, Yi Liu, Zun Wang, Jilan Xu, Guo Chen, Ping Luo, et al. 2024. Mvbench: A comprehensive multi-modal video understanding benchmark. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. IEEE, 22195–22206.
- [20] Qi Li, Tzu-Chen Chiu, Hsiang-Wei Huang, Min-Te Sun, and Wei-Shinn Ku. 2024. Videobadminton: a video dataset for badminton action recognition. In *2024 IEEE International Conference on Big Data*. IEEE, 1387–1392.
- [21] Xiaojie Li, Shaowei He, Jianlong Wu, Yue Yu, Liqiang Nie, and Min Zhang. 2023. Mask again: Masked knowledge distillation for masked video modeling. In *Proceedings of the ACM International Conference on Multimedia*. ACM, 2221–2232.
- [22] Xiaojie Li, Jianlong Wu, Shaowei He, Shuo Kang, Yue Yu, Liqiang Nie, and Min Zhang. 2023. Fine-grained key-value memory enhanced predictor for video representation learning. In *Proceedings of the ACM International Conference on Multimedia*. ACM, 2264–2274.
- [23] Yixuan Li, Lei Chen, Runyu He, Zhenzhi Wang, Gangshan Wu, and Limin Wang. 2021. Multisports: A multi-person video dataset of spatio-temporally localized sports actions. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 13536–13545.
- [24] Zijiang Liang, Yanjie Xu, Yifan Hong, Penghui Shang, Qi Wang, Qiang Fu, and Ke Liu. 2024. A Survey of Multimodal Large Language Models. In *Proceedings of International Conference on Computer, Artificial Intelligence and Control Engineering*. 405–409.
- [25] Zhaoyu Liu, Kan Jiang, Murong Ma, Zhe Hou, Yun Lin, and Jin Song Dong. 2025. F3Set: Towards Analyzing Fast, Frequent, and Fine-grained Events from Videos. *arXiv preprint arXiv:2504.08222* (2025).
- [26] Karttikeya Mangalam, Raiymbek Akshulakov, and Jitendra Malik. 2023. Egoschema: A diagnostic benchmark for very long-form video language understanding. In *Proceedings of the Advances in Neural Information Processing Systems*, Vol. 36. 46212–46244.
- [27] OpenAI. 2025. GPT-4.1. <https://openai.com/index/gpt-4-1/>
- [28] Jiayuan Rao, Haoning Wu, Hao Jiang, Ya Zhang, Yanfeng Wang, and Weidi Xie. 2025. Towards Universal Soccer Video Understanding. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. IEEE.
- [29] Jiayuan Rao, Haoning Wu, Chang Liu, Yanfeng Wang, and Weidi Xie. 2024. MatchTime: Towards Automatic Soccer Game Commentary Generation. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*.
- [30] Dian Shao, Yue Zhao, Bo Dai, and Dahua Lin. 2020. Finegym: A hierarchical video dataset for fine-grained action understanding. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. IEEE, 2616–2625.
- [31] Yansong Tang, Dajun Ding, Yongming Rao, Yu Zheng, Danyang Zhang, Lili Zhao, Jiwen Lu, and Jie Zhou. 2019. Coin: A large-scale dataset for comprehensive instructional video analysis. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. IEEE, 1207–1216.
- [32] Yunlong Tang, Junjia Guo, Hang Hua, Susan Liang, Mingqian Feng, Xinyang Li, Rui Mao, Chao Huang, Jing Bi, Zeliang Zhang, et al. 2025. Vidcomposition: Can mllms analyze compositions in compiled videos?. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 8490–8500.
- [33] ByteDance Seed Team. 2025. Doubao-1.5-pro. https://seed.bytedance.com/en/special/doubao_1_5_pro
- [34] Gemini Team, Rohan Anil, Sebastian Borgeaud, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew M Dai, Anja Hauth, Katie Millican, et al. 2023. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805* (2023).
- [35] Zhan Tong, Yibing Song, Jue Wang, and Limin Wang. 2022. VideoMAE: Masked Autoencoders are Data-Efficient Learners for Self-Supervised Video Pre-Training. In *Proceedings of the Advances in Neural Information Processing Systems*. 10078–10093.
- [36] Limin Wang, Bingkun Huang, Zhiyu Zhao, Zhan Tong, Yanan He, Yi Wang, Yali Wang, and Yu Qiao. 2023. VideoMAE V2: Scaling Video Masked Autoencoders with Dual Masking. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. IEEE, 14549–14560.
- [37] Wei-Yao Wang, Wei-Wei Du, Wen-Chih Peng, and Tsi-Ui Ik. 2024. Benchmarking Stroke Forecasting with Stroke-Level Badminton Dataset. In *Proceedings of International Joint Conference on Artificial Intelligence*. 8829–8832.
- [38] Wei-Yao Wang, Yung-Chang Huang, Tsi-Ui Ik, and Wen-Chih Peng. 2023. Shut-tleset: A human-annotated stroke-level singles dataset for badminton tactical analysis. In *Proceedings of ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 5126–5136.
- [39] Xiao Wang, Jingyun Hua, Weihong Lin, Yuanxing Zhang, Fuzheng Zhang, Jianlong Wu, Di Zhang, and Liqiang Nie. 2025. HAIC: Improving Human Action Understanding and Generation with Better Captions for Multi-modal Large Language Models. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*. ACL, 10158–10181.
- [40] Xiao Wang, Qingyi Si, Shiyu Zhu, Jianlong Wu, Li Cao, and Liqiang Nie. 2025. AdaReTaKe: Adaptive Redundancy Reduction to Perceive Longer for Video-language Understanding. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*. ACL, 5417–5432.

- [41] Xiao Wang, Jianlong Wu, Zijia Lin, Fuzheng Zhang, Di Zhang, and Liqiang Nie. 2025. Video DataFlywheel: Resolving the Impossible Data Trinity in Video-Language Understanding. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2025), 2912–2923.
- [42] Tao Wu, Runyu He, Gangshan Wu, and Limin Wang. 2024. Sportshhi: A dataset for human-human interaction detection in sports videos. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. IEEE, 18537–18546.
- [43] Haotian Xia, Zhengbang Yang, Junbo Zou, Rhys Tracy, Yuqing Wang, Chi Lu, Christopher Lai, Yanjun He, Xun Shao, Zhuoqing Xie, Yuan fang Wang, Weining Shen, and Hanjie Chen. 2025. SPORTU: A Comprehensive Sports Understanding Benchmark for Multimodal Large Language Models. In *Proceedings of the International Conference on Learning Representations*.
- [44] Dejing Xu, Zhou Zhao, Jun Xiao, Fei Wu, Hanwang Zhang, Xiangnan He, and Yueting Zhuang. 2017. Video Question Answering via Gradually Refined Attention over Appearance and Motion. In *Proceedings of the ACM International Conference on Multimedia*. ACM, 1645–1653.
- [45] Jinglin Xu, Guohao Zhao, Sibao Yin, Wenhao Zhou, and Yuxin Peng. 2024. Finesports: A multi-person hierarchical sports video dataset for fine-grained action understanding. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. IEEE, 21773–21782.
- [46] Dongjie Yang, Suyuan Huang, Chengqiang Lu, Xiaodong Han, Haoxin Zhang, Yan Gao, Yao Hu, and Hai Zhao. 2024. Vript: A video is worth thousands of words. In *Proceedings of the Advances in Neural Information Processing Systems*, Vol. 37. 57240–57261.
- [47] Boqiang Zhang, Kehan Li, Zesen Cheng, Zhiqiang Hu, Yuqian Yuan, Guanzheng Chen, Sicong Leng, Yuming Jiang, Hang Zhang, Xin Li, et al. 2025. VideoLLaMA 3: Frontier Multimodal Foundation Models for Image and Video Understanding. *arXiv preprint arXiv:2501.13106* (2025).
- [48] Shiyi Zhang, Wenxun Dai, Sujia Wang, Xiangwei Shen, Jiwen Lu, Jie Zhou, and Yansong Tang. 2023. Logo: A long-form video dataset for group action quality assessment. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. IEEE, 2405–2414.

A The FineBadminton

We introduce FineBadminton, a comprehensive, and multi-modal dataset designed for fine-grained analysis of badminton matches. As shown in the comparative analysis in Table 3, while numerous datasets for sports video analysis exist, there is a notable gap, particularly in the realm of badminton. Previous badminton datasets like ShuttleNet, ShuttleSet, and ShuttleSet22 have primarily focused on stroke detection and rally localization, lacking the detailed, multi-faceted annotations necessary for deeper semantic understanding.

Table 3: Comparison of sports video analysis datasets. Our proposed dataset, FineBadminton, is highlighted in blue. Abbreviations: HA (Hierarchical Annotation), TA (Textual Annotation), UED (Uses Existing Data, e.g., scores, commentary), AQA (Action Quality Assessment), FGL (Fine-grained Labels).

Dataset	Sport	Videos	Rallies/Clips	Strokes/Actions/Evts.	HA	TA	UED	AQA	FGL
Finegym(CVPR'20)	Gymnastics	303	4,883	32,697	✓	✗	✓	✓	✓
SoccerNetV2(CVPR'21)	Soccer	500	-	110,458	✓	✗	✗	✗	✓
FineDiving(CVPR'22)	Diving	135	-	3,000	✗	✗	✗	✓	✓
ShuttleNet(AAAI'22)	Badminton	75	4,325	43,191	✗	✗	✗	✗	✗
ShuttleSet(KDD'23)	Badminton	44	3,685	36,492	✗	✗	✗	✗	✗
ShuttleSet22(IJCAI'23)	Badminton	58	3,992	33,612	✗	✗	✗	✗	✗
Logo(CVPR'23)	Artistic Swim.	200	-	15,764	✓	✗	✓	✓	✓
FineSports(CVPR'24)	Basketball	-	10,000	16,000	✓	✗	✗	✗	✓
P2anet(TOMM'24)	Table Tennis	200	2,721	139,075	✗	✗	✗	✗	✗
SportsHHI(CVPR'24)	B-ball/V-ball	160	-	50,649	✓	✗	✗	✗	✓
F ³ Set(ICLR'25)	various	114	11,584	42,846	✓	✗	✗	✗	✓
FineBadminton	Badminton	120	3,215	33,325	✓	✓	✗	✓	✓

B The FBBench

B.1 Evaluation Protocol and Prompting

The evaluation protocol for the FBBench is tailored to its two distinct question formats. For the multiple-choice tasks, models are instructed to respond with a single letter corresponding to their chosen option. A structured prompt is employed: "This is a segment from a badminton match. 'top' refers to the player at the top of the screen, and 'bottom' refers to the player at the bottom. {Question} Watch the video carefully, paying attention to the rally's progression, and the players' actions and postures. Based on your observation, select the option that most accurately answers the question. Respond with a single option letter." Robust answer extraction is ensured through predefined templates, supplemented by GLM-4-air[9] for non-standard responses. To enhance rigor and mitigate guessing, a re-evaluation loop with shuffled options is implemented for initially correct answers.

B.2 Questions Mapping

Table 4: Mapping of Spatial and Temporal Dimensions to Task IDs. This table outlines the different categories and their corresponding identifiers used in our analysis.

Spatial Dimension	Temporal Dimension	ID
Count	Hitting Count	T1
	Round Count	T2
	Detailed Hitting Count	T3
Action	Action Prediction	T4
	Action Grounding	T5
	Action Classification	T6
Position	Moving Recognition	T7
	Hitting Localization	T8
	Landing Prediction	T9
Cognition	Hitting Comment	T10
	Round Comment	T11
	Pointer Recognition	T12